# SoccerGMR SI-Bind: Detailed Component Ablation Study (E1 to E7)

Fine-grained ablation across mechanisms: Mechanism A (Binding & Residual), Mechanism B (Slot Innovation), and Mechanism C (Dominance Refresh) on Standard Test (1,036 queries) and Full Test (2,970 queries).

| | | Standard Test Split (1,036 queries) | | | | Full-Decodable Test Split (2,970 queries) | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Backbone** | **Ablation Method / Configuration** | **mAP (%)** | **Orc-Gap (%)** | **Rej-F1@.5 (%)** | **AUROC (%)** | **mAP (%)** | **Orc-Gap (%)** | **Rej-F1@.5 (%)** | **AUROC (%)** |
| **Moment-DETR** | 1. E1: Binding-only (Interlayer Anchor) | 12.32 | **18.82** | **75.18** | **75.96** | 10.34 | **17.98** | 78.49 | **82.38** |
| | 2. E2: Lite Binding + Residual Stream | 11.85 | 22.41 | 72.40 | 71.55 | 10.14 | 21.20 | 76.11 | 80.14 |
| | **3. E3: E2 + Slot-Private Innovation (B)** | **15.60** | 24.26 | 66.22 | 73.17 | **13.30** | 23.02 | 67.12 | 80.91 |
| | 4. E4: E2 + Uniform Random Refresh (C Control) | 12.35 | 24.02 | 60.18 | 75.53 | 12.13 | 22.83 | 61.91 | 81.77 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 7.95 | 21.56 | 75.00 | 68.32 | 6.47 | 21.06 | **78.77** | 77.80 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 15.21 | 23.66 | 65.52 | 74.32 | 12.58 | 22.21 | 68.83 | 81.86 |
| | 7. E7: E6 - Binding (bind_coef=0 Ablation) | 15.21 | 23.66 | 65.52 | 74.32 | 12.58 | 22.21 | 68.83 | 81.86 |
| **BM-DETR** | 2. E2: Lite Binding + Residual Stream | 6.36 | 18.84 | **75.00** | 68.63 | 5.68 | 18.46 | 78.80 | 78.79 |
| | **3. E3: E2 + Slot-Private Innovation (B)** | **7.88** | 21.63 | 72.74 | **71.55** | **7.08** | 20.95 | 76.91 | **79.45** |
| | 4. E4: E2 + Uniform Random Refresh (C Control) | 6.48 | 20.60 | **75.00** | 69.37 | 5.95 | 19.98 | **78.91** | 79.18 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 7.16 | **15.26** | **75.00** | 69.37 | 6.93 | **15.16** | 78.14 | 78.92 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 7.72 | 20.91 | 73.40 | 70.18 | 7.03 | 20.30 | 77.47 | 79.27 |
| | 7. E7: E6 - Binding (bind_coef=0 Ablation) | 7.72 | 20.91 | 73.40 | 70.18 | 7.03 | 20.30 | 77.47 | 79.27 |
| **UVCOM** | 1. E1: Binding-only (Interlayer Anchor) | 15.31 | 18.42 | 74.94 | 77.68 | 14.60 | 17.55 | 73.74 | 78.58 |
| | 2. E2: Lite Binding + Residual Stream | 20.31 | **14.46** | 72.79 | 77.04 | 20.02 | **14.54** | 70.95 | 76.70 |
| | **3. E3: E2 + Slot-Private Innovation (B)** | **20.87** | 18.11 | 69.84 | 77.23 | **20.95** | 16.99 | 69.72 | 76.14 |
| | 4. E4: E2 + Uniform Random Refresh (C Control) | 17.30 | 16.39 | 74.17 | 77.79 | 16.34 | 15.66 | **73.78** | **78.94** |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 20.49 | 17.55 | 69.48 | **78.13** | 19.76 | 16.74 | 68.12 | 75.69 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 18.99 | 16.30 | **75.40** | 76.91 | 17.74 | 15.55 | 73.11 | 75.20 |
| | 7. E7: E6 - Binding (bind_coef=0 Ablation) | | | -- | | | | -- | |
| **CG-DETR** | 2. E2: Lite Binding + Residual Stream | 14.99 | 23.42 | 72.77 | 73.86 | 12.68 | 22.41 | 75.71 | **81.67** |
| | **3. E3: E2 + Slot-Private Innovation (B)** | 16.29 | 24.97 | 64.69 | 75.17 | 14.54 | 23.68 | 65.94 | 80.57 |
| | 4. E4: E2 + Uniform Random Refresh (C Control) | 14.47 | **20.87** | 75.19 | 74.40 | 13.00 | **19.67** | **78.33** | 81.20 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 13.80 | 21.77 | **75.72** | 75.42 | 13.96 | 21.14 | 77.43 | 81.33 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | **17.38** | 22.89 | 70.85 | **75.92** | **15.54** | 21.61 | 73.14 | 80.94 |
| | 7. E7: E6 - Binding (bind_coef=0 Ablation) | | | -- | | | | -- | |
| **QD-DETR** | **6. E6: E2 + Innovation (B) + Dominance (C)** | 16.30 | 20.65 | 73.71 | 76.63 | 13.20 | 19.84 | 75.91 | 82.47 |

Note: E1 = Binding only (identity across layers); E2 = Lite Binding + Residual; E3 = E2 + Slot-Private Innovation (B); E4 = E2 + Uniform Random Refresh (C Control); E5 = E2 + Dominance Refresh (C); E6 = E2+B+C (Champion Triad); E7 = E6 - Binding.
Bold numbers indicate best performance among ablation variants within the same backbone. Oracle Gap: lower is better.